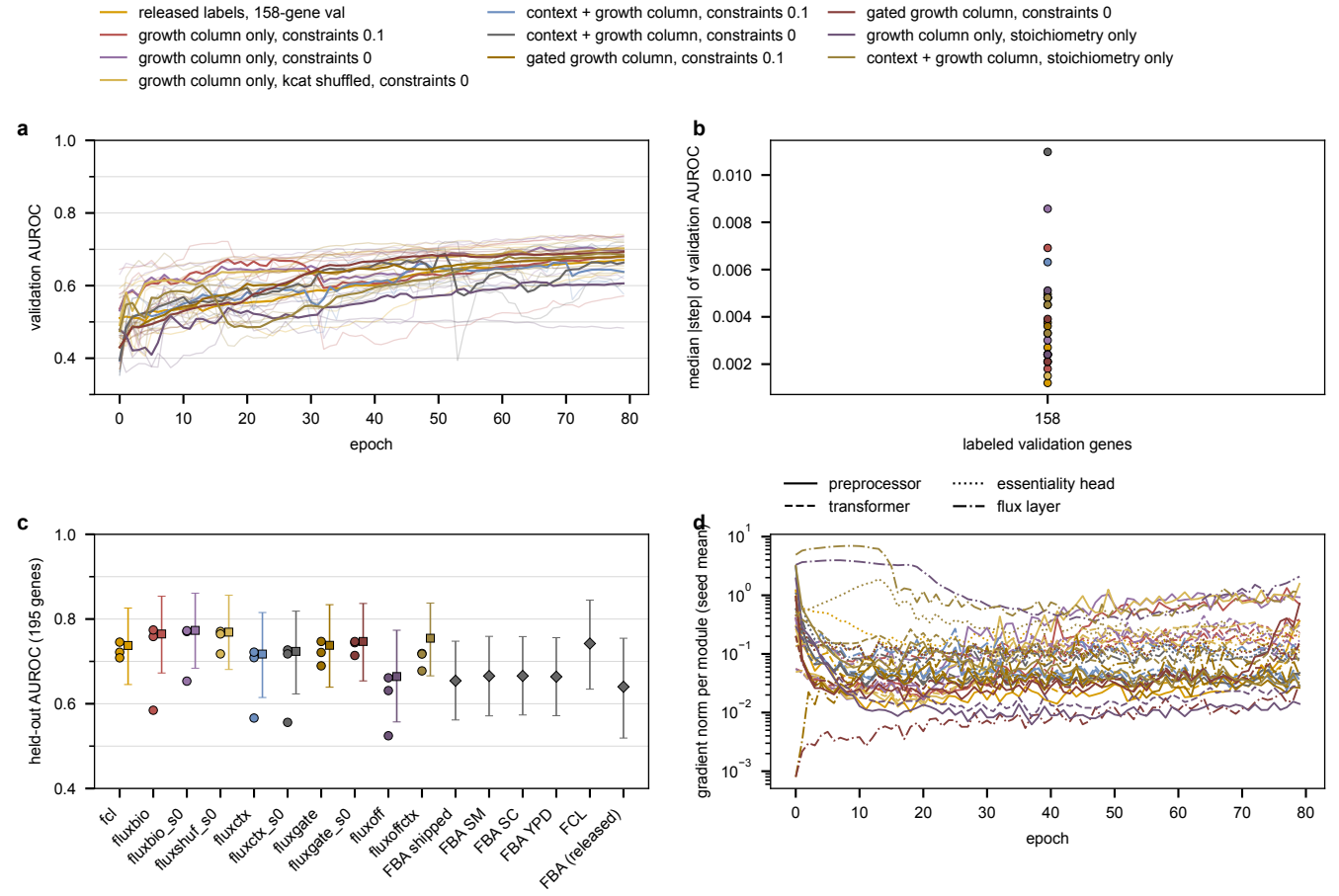

released labels, 158-gene val
growth column only, constraints 0.1
growth column only, constraints 0
growth column only, kcat shuffled, constraints 0
context + growth column, constraints 0.1
context + growth column, constraints 0
gated growth column, constraints 0.1
gated growth column, constraints 0
growth column only, stoichiometry only
context + growth column, stoichiometry only
a
validation AUROC
epoch
b
median |step| of validation AUROC
labeled validation genes
158
c
held-out AUROC (195 genes)
fcl
fluxbio
fluxbio_s0
fluxshuf_s0
fluxctx
fluxctx_s0
fluxgate
fluxgate_s0
fluxoff
fluxoffctx
FBA shipped
FBA SM
FBA SC
FBA YPD
FCL
FBA (released)
d
preprocessor
transformer
essentiality head
flux layer
gradient norm per module (seed mean)
epoch